Script Before Sound: Parameter-Efficient Adaptation of Whisper for Low-Resource Sindhi Speech Recognition

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Abstract

Automatic speech recognition for Sindhi, a language with approximately 30 million speakers, remains poorly supported by general-purpose multilingual models. This study examines the behaviour of OpenAI's Whisper on Sindhi and evaluates strategies for improving it. When forced to decode in Sindhi, the unadapted whisper-small model produced output in the Perso-Arabic script for only 3% of 328 validated Mozilla Common Voice utterances; the remainder was distributed across Devanagari, romanized Latin, Telugu and Khmer characters, yielding a word error rate (WER) of 142.5%. Low-Rank Adaptation (LoRA) fine-tuning was subsequently applied, and nine modifications were evaluated sequentially, each retained only if it reduced WER on a held-out development set. Fine-tuning alone resolved the script confusion. Beam search decoding and the extension of LoRA adapters from the attention query and value projections to all linear layers produced further gains, whereas label normalization, data filtering, n-gram repetition blocking, speed perturbation and extended training did not. Trained on 2,000 utterances, whisper-small attained 18.7% WER and 6.4% character error rate (CER), with no evidence of saturation. Among model sizes trained on 500 utterances, whisper-large-v3-turbo performed best (25.9% WER); however, whisper-small trained on twice as much data achieved a lower error rate (23.0%). A rule-based analysis of all 328 test transcriptions indicates that approximately two-thirds of the residual errors are misspellings of one or two characters. Two evaluation pitfalls specific to Sindhi are also documented: conventional Unicode normalization removes the frequent function words ۾ ('in') and ۽ ('and'), and 1,351 candidate training utterances duplicated test-set sentences and were therefore excluded to prevent leakage.

Keywords: automatic speech recognition, Sindhi, low-resource languages, Whisper, parameter-efficient fine-tuning, evaluation methodology

1. Introduction

Sindhi is an Indo-Aryan language spoken by approximately 30 million people, principally in the Sindh province of Pakistan and among diaspora communities in India. Despite this substantial speaker population, the language has very limited transcribed speech resources. Whisper [6], a widely adopted multilingual speech recognition model, lists Sindhi among its supported languages, yet its performance on Sindhi is markedly weaker than on high-resource languages. In practical terms, voice interfaces, automatic captioning and dictation remain largely unavailable to Sindhi speakers.

Sindhi additionally presents a writing-system challenge. In Pakistan it is written predominantly in an extended Perso-Arabic alphabet, whereas some speakers in India employ the Devanagari script. The Perso-Arabic orthography contains letters absent from Urdu and Persian, including the implosive consonants ٻ, ڄ, ڃ and ڏ. Consequently, a recognizer may perceive Sindhi speech accurately and nevertheless fail entirely if it renders its output in an inappropriate script.

A preliminary phase of this project fine-tuned whisper-medium on 200 utterances and reported a reduction in WER from 148.35% to 72.97% under unnormalized scoring. Subsequent review of that work raised two questions: how Whisper models of different sizes compare on Sindhi, and which language-specific design choices produce measurable improvements. The present study addresses both questions under a more rigorous evaluation protocol. Its principal contributions are as follows:

1. A diagnosis of the failure mode of unadapted Whisper on Sindhi, based on an inventory of the characters it produces.
2. A controlled sequence of nine modifications, each retained or rejected according to its effect on an independent development set.
3. A data-scaling analysis spanning 100 to 2,000 training utterances.
4. A comparison of whisper-small, whisper-medium and whisper-large-v3-turbo in both zero-shot and fine-tuned settings.
5. An evaluation protocol for Sindhi comprising sentence-disjoint data splits, a normalizer that preserves ۾ and ۽, and reporting of WER and CER at four normalization levels.

2. Related work

Whisper. Radford et al. [6] trained Whisper on 680,000 hours of weakly supervised audio spanning 99 languages. Their analysis shows that WER for a given language decreases steadily with the quantity of training audio available for it, which accounts for the weak performance observed on languages such as Sindhi. The same work emphasizes that text normalization exerts a substantial influence on reported WER.

Parameter-efficient fine-tuning. Low-Rank Adaptation (LoRA) [2] freezes the weights of a pretrained model and trains small low-rank matrices inserted alongside selected layers. The resulting reduction in memory requirements permits Whisper to be fine-tuned on a single free-tier

graphics processing unit (GPU), the setting in which the present study was conducted. The implementation relies on the Transformers [7] and PEFT [4] libraries.

Data and augmentation. Common Voice [1] is a crowd-sourced corpus in which volunteers record prompted sentences and other contributors validate each recording by vote. Because many speakers read the same sentence, the corpus carries a risk of train-test leakage, which is addressed in Section 3. Speed perturbation [3] is an established augmentation technique that resamples audio at slightly altered rates.

Sindhi speech recognition. To the author's knowledge, controlled evaluations of Whisper on Sindhi are scarce, and no prior work known to the author examines the script confusion or normalization issues reported in this paper.

3. Data and evaluation protocol

All experiments employ the Sindhi portion (locale sd) of Mozilla Common Voice Scripted Speech 27.0, obtained from the Mozilla Data Collective. The release comprises 328 community-validated utterances and 65,569 utterances that have not yet been reviewed. The partition of these data is summarized in Table 1, and the complete experimental pipeline is illustrated in Figure 1.

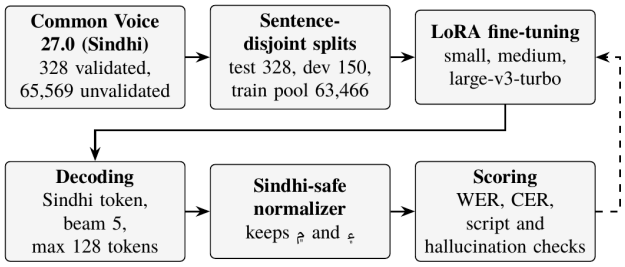


Figure 1. Experimental pipeline. The dashed arrow denotes the retention rule: a modification is retained only if development-set WER decreases by at least 0.5 points. The test set is used exclusively for final reporting.

Table 1. Data partitions.

Split	Source	Count	Role
Test	Validated	328	Final reporting only
Development	Unvalidated	150	All retention decisions
Training pool	Unvalidated	63,466	Nested subsets of 100 to 2,000

Sentence-disjoint partitioning. Common Voice contributors read from a shared pool of prompts, so an identical sentence frequently occurs many times with different speakers. To prevent a model from partially memorizing test sentences through recordings by other speakers, 1,351 pool utterances whose normalized transcript matched a test sentence were removed, together with 602 utterances matching a development sentence.

Nested subsets. The training pool was shuffled once with a fixed seed (42), and a run with N utterances uses the first N items, so that each larger subset contains every smaller one. An optional quality filter excludes utterances with more negative than positive votes and transcripts containing Latin letters, digits or Devanagari characters; it retains 62,673 of the 63,466 pool utterances.

Metrics. WER and CER were computed with the jiwer library, following the measures described by Morris et al. [5], at the four normalization levels defined in Table 2. The *full* level maps Persian ی, Arabic ى and Urdu ے to Sindhi ي, Arabic ك to Sindhi ڪ, and Urdu ہ to ه. It intentionally does not merge ک and ڪ, which denote distinct phonemes in Sindhi (/k^h/ and /k/ respectively).

Table 2. Text-normalization levels. Each level includes all operations of the preceding level.

Level	Operations
raw	Trimming and collapsing of whitespace; identical to the preliminary study
punct	NFKC normalization; removal of punctuation, invisible characters and tatweel
full	Removal of diacritics, digit conversion and unification of encoding variants of the same letter (headline metric)
lenient	Mapping of Urdu letters to the Sindhi letters representing the same sound, e.g. ٹ to ٽ

A Sindhi-specific normalization hazard. The Unicode standard assigns ۾ (U+06FE, 'in') and ۽ (U+06FD, 'and') to the symbol category rather than the letter category. Any normalizer that removes all punctuation and symbol characters therefore deletes these words prior to scoring; Whisper's reference text normalizer behaves in precisely this manner. The initial normalizer developed for this study contained the same defect, which distorted all scores until it was corrected. Every result reported here uses the corrected normalizer.

Failure diagnostics. For each run, two additional quantities were recorded: the proportion of outputs written predominantly in Perso-Arabic script, and the proportion of hallucinated outputs. An output was classified as hallucinated if it contained a character repeated eight or more times, the same word four times consecutively, or more than three times the reference length plus 20 characters. Empty outputs were scored as deletion of every reference word.

4. Methods

Models. Three publicly released checkpoints were evaluated: whisper-small (244 million parameters), whisper-medium (769 million) and whisper-large-v3-turbo (809 million), the last being a variant of large-v3 with a reduced decoder.

Fine-tuning. All base weights were frozen, and LoRA adapters were trained with rank 32, scaling factor $\alpha = 64$ and dropout 0.05. The original recipe adapted only the query and value projections of the attention layers. Optimization used AdamW with a learning rate of 0.001 and weight decay of 0.01, a linear schedule with 10% warm-up, five epochs, an effective batch size of 16, 16-bit mixed precision and gradient clipping at 1.0. Target

transcripts were prefixed with Whisper's Sindhi language and transcription task tokens.

Decoding. Unless otherwise stated, decoding forced the Sindhi language token, used the transcription task and limited output to 128 new tokens. Greedy decoding and beam search with five beams were compared within the ablation sequence.

Computational environment. Each run used a single NVIDIA T4 GPU on the free tiers of Google Colab and Kaggle. A fixed seed (42) governed sampling, shuffling and initialization.

Sequential ablation. Beginning with the original recipe applied to whisper-small with 500 training utterances, each step altered a single setting relative to the best configuration identified so far. A modification was retained only if it reduced development-set WER (full normalization) by at least 0.5 percentage points. The steps are listed in Table 3.

Table 3. Ablation steps and their motivation.

Step	Modification	Motivation
L0	Unadapted model, automatic language identification	Baseline behaviour
L1	Forced Sindhi language token	Prevent language drift
L2	LoRA fine-tuning, original recipe	Adaptation to Sindhi
L3	Beam search (5), with and without 3-gram blocking	Reduce repetitive output
L4	Training on normalized transcripts	Label consistency
L5	Quality-filtered training data	Remove poorly rated clips
L6	LoRA on all linear layers (q, k, v, out, fc1, fc2)	Greater adaptation capacity
L7	Speed perturbation (0.9×, 1.1×)	Speaking-rate variability
L8	Ten epochs instead of five	Test for under-training

Scaling and model-size experiments. The best configuration was trained on whisper-small with 100, 250, 500, 1,000 and 2,000 utterances. Each model size was evaluated in a zero-shot setting (Sindhi token forced, beam search) and after fine-tuning with the best configuration on 500 utterances. In addition, the recipe of the preliminary study (whisper-medium, 200 utterances, query and value LoRA, greedy decoding) was reproduced under the revised protocol.

5. Results

5.1 Script confusion in the unadapted model

Forcing the Sindhi language token degraded, rather than improved, the performance of the unadapted whisper-small model (Table 4). With automatic language identification the model attained 112.3% WER; when instructed that the input was Sindhi, WER rose to 142.5%, and only 3% of outputs were written in the Perso-Arabic script.

Table 4. Unadapted and fine-tuned whisper-small on the test set (WER and CER in %). PA denotes the proportion of outputs in Perso-Arabic script; Hall. denotes the proportion of hallucinated outputs.

Configuration	WER	CER	PA	Hall.
Automatic language (L0)	112.3	90.0	26%	6%
Sindhi forced (L1)	142.5	124.3	3%	27%
Fine-tuned, 500 utt. (L2)	41.4	14.3	100%	0%

An inventory of output characters clarifies the nature of this failure. Table 5 reports the most frequent characters in the L1 output that occur in none of the 328 reference transcripts. Unaccented Latin letters were also frequent (k: 296 occurrences; y: 277; o: 269). The only character reflecting a plausible near-confusion is ڏ, the Urdu letter for the retroflex stop that Sindhi writes as ڊ. After fine-tuning, the same analysis identified a single character outside the Sindhi alphabet, one Latin "y", across all 328 outputs.

Table 5. Most frequent characters in the output of unadapted Whisper in Sindhi mode that are absent from all references.

Char.	Code point	Script	Count
◌្	U+17D2	Khmer	697
ḍ	U+1E0D	Latin transliteration	367
◌ీ	U+0C3F	Telugu	260
ౙ	U+0C28	Telugu	209
ڏ	U+0688	Urdu	186
Ḥ	U+1E24	Latin transliteration	169
◌ॅ	U+094D	Devanagari	134
ὸ	U+1F40	Greek	42

A plausible explanation is that Whisper's training data contained very little text written in Sindhi, so that its Sindhi language token is only weakly associated with the Perso-Arabic script. This hypothesis cannot be verified directly, since the training data have not been released. Beam search further degraded the unadapted model: zero-shot whisper-small reached 192.3% WER with five beams, compared with 142.5% under greedy decoding.

5.2 Sequential ablation

Three modifications satisfied the retention criterion: LoRA fine-tuning, beam search, and the application of LoRA to all linear layers (Table 6).

Table 6. Sequential ablation on whisper-small with 500 training utterances (WER and CER in %, full normalization; Raw WER uses unnormalized scoring). All fine-tuned runs produced 100% Perso-Arabic output and no hallucinations.

Step	Modification	Dev WER	Test WER	Test CER	Raw WER	Decision
L0	Unadapted, automatic language	110.3	112.3	90.0	115.8	Baseline
L1	Forced Sindhi token	148.1	142.5	124.3	143.7	Reference
L2	LoRA fine-tuning	40.0	41.4	14.3	47.5	Retained
L3	Beam search (5)	37.9	39.7	13.5	45.6	Retained
L3	Beam search with 3-gram blocking	41.0	42.5	14.6	48.2	Rejected
L4	Normalized training transcripts	38.2	39.1	13.7	45.5	Rejected
L5	Quality-filtered data	37.8	39.1	13.5	45.0	Rejected
L6	LoRA on all linear layers	31.0	**32.0**	**11.0**	38.5	Retained
L7	Speed perturbation	31.3	30.2	10.6	35.8	Rejected

Step	Modification	Dev WER	Test WER	Test CER	Raw WER	Decision
L8	Ten epochs	33.1	31.1	10.9	36.9	Rejected

Fine-tuning accounted for the largest improvement, reducing test WER from 142.5% to 41.4% and eliminating the script confusion. Beam search lowered development WER by a further 2.1 percentage points. Extending LoRA from the query and value projections to every linear layer produced the largest gain beyond fine-tuning itself, reducing WER by 6.9 points on the development set and 7.8 points on the test set. Taken together, the retained modifications reduced test WER from 41.4% to 32.0% with the amount of training data held constant.

Several other modifications produced no measurable benefit. Normalizing training transcripts and filtering utterances changed development WER by less than 0.5 points; the quality filter replaced only 6 of the 500 training utterances, which suggests that the Common Voice Sindhi transcripts are already largely consistent. Blocking repeated 3-grams increased error, most likely because it also suppressed legitimate repetitions of subword units. Training for ten epochs resulted in overfitting.

Speed perturbation yielded the only ambiguous outcome. It was marginally worse on the development set (31.3% versus 31.0%) but noticeably better on the test set (30.2% versus 32.0%). In accordance with the protocol it was rejected; the discrepancy is interpreted as evidence that differences of one to two points lie within the sampling variability of a 150-utterance development set.

5.3 Effect of training-data quantity

Between 100 and 2,000 training utterances, test WER decreased from 55.3% to 18.7% and CER from 21.0% to 6.4% (Figure 2). Although the marginal gains diminish (9.0 points between 500 and 1,000 utterances and 4.3 points between 1,000 and 2,000), the curve shows no sign of saturation, and more than 60,000 unused utterances remain available. No hallucinated output was observed at any training size.

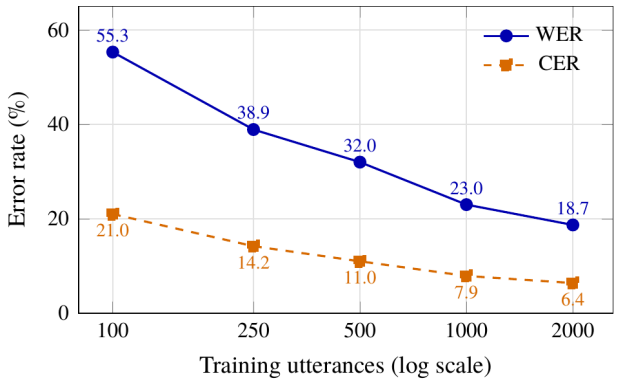


Figure 2. Test-set WER and CER of whisper-small under the best configuration as a function of training-data quantity. The unadapted model attains 142.5% WER.

5.4 Effect of model size

Larger models exhibit lower zero-shot error and modestly lower error after fine-tuning (Table 7, Figure 3). The transition from whisper-small to whisper-medium reduced fine-tuned WER by 4.9 points, whereas the transition from whisper-medium to whisper-large-v3-turbo reduced it by only 1.2 points, accompanied by a further 1.7-point reduction in CER. Training-data quantity proved more influential than model size: whisper-small trained on 1,000 utterances (23.0% WER) outperformed whisper-large-v3-turbo trained on 500 utterances (25.9%) despite having fewer than one third of its parameters.

Table 7. Comparison of model sizes on the test set (WER and CER in %). Zero-shot (ZS) results use the forced Sindhi token and beam search; fine-tuned (FT) results use the best configuration trained on the same 500 utterances. Time is training time in minutes.

Model	Params	ZS WER	ZS CER	FT WER	FT CER	Time (min)
small	244M	192.3	153.5	32.0	11.0	10
medium	769M	150.9	116.8	27.1	10.4	26
large-v3-turbo	809M	108.9	98.2	**25.9**	**8.7**	29

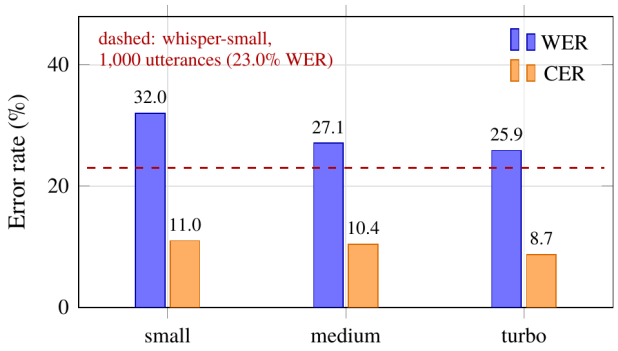


Figure 3. Fine-tuned error rates by model size with 500 training utterances. The dashed line indicates the WER of whisper-small trained on 1,000 utterances.

When the recipe of the preliminary study (whisper-medium, 200 utterances, query and value LoRA, greedy decoding) was reproduced under the revised protocol, it attained 44.0% WER and 16.1% CER. The preliminary study had reported 72.97% on an earlier release using unnormalized scoring and unbounded decoding; the difference is therefore attributable largely to evaluation choices rather than to the model itself. Under the revised configuration with 500 utterances, whisper-medium attained 27.1% WER, although this improvement conflates the change of recipe with a 2.5-fold increase in training data. The effect of the recipe alone is most clearly isolated on whisper-small at a fixed data quantity, where WER fell from 41.4% to 32.0%.

5.5 Influence of text normalization

For the fine-tuned model, removal of punctuation alone reduced WER by 6.3 points (Table 8). The letter-variant mappings contributed only a further 0.2 points, since the fine-tuned model already produces standard Sindhi orthography. For the unadapted model, normalization had a

negligible effect, as no character mapping can recover Sindhi text from output written in unrelated scripts.

Table 8. WER and CER (%) of identical transcriptions under each normalization level.

Model	WER raw	WER punct	WER full	WER len.	CER raw	CER full
Unadapted (L1)	143.7	142.6	142.5	142.4	124.7	124.3
Fine-tuned (L6)	38.5	32.2	32.0	32.0	12.8	11.0

A further orthographic variant, the Urdu-style letter ھ, occurs in 5 of the 328 references but never in model output. Treating it as equivalent to ه would reduce the error of the best model from 18.70% to 18.50% WER and from 6.38% to 6.31% CER. The headline results are reported without this mapping.

5.6 Analysis of residual errors

Each test transcription produced by the best model (whisper-small trained on 2,000 utterances) was aligned with its reference, and every mismatched region was classified by deterministic rules. A single word differing by at most two characters was classified as a near-miss. A region whose characters match once spaces are removed, or match closely with a different word count, was classified as a word-boundary error. Missing and inserted words were counted directly, a reference containing Latin characters was classified as faulty, and all remaining regions were classified as substitutions of a different word.

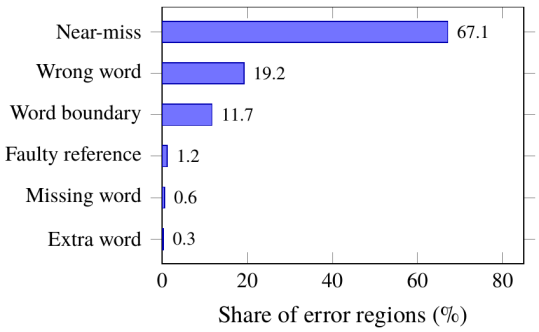


Figure 4. Distribution of 343 error regions across all 328 test utterances for the best model, as classified by deterministic rules. A near-miss denotes a single word differing by one or two characters.

After normalization, 30.5% of test utterances (100 of 328) were transcribed without error, and 80.5% were either error-free or contained only near-misses and word-boundary errors (Figure 4). Deletions and insertions of whole words were rare. A substantial share of whole-word substitutions involved proper nouns and loanwords; for example, ملائيشيا (Malaysia) was transcribed as ملايشه, and آسٽريليا (Australia) as حسليا. Three references contained Latin characters, including one in which the model produced خاتون for a reference recorded as "Khatun".

These categories should be interpreted with caution, as the rules are necessarily coarse. For instance, the transcription of تون ('you') as هو ('he') is classified as a near-miss because only two characters differ, even though the meaning is altered.

6. Discussion

Orthography precedes acoustics. The central finding of this study is that the deficiency of unadapted Whisper on Sindhi originates in orthography rather than in acoustic modelling. A model that renders Sindhi speech in Khmer diacritics cannot be corrected through improved decoding or scoring alone. Fine-tuning on 500 utterances constrained every output to the correct alphabet, and as few as 100 utterances, amounting to only a few minutes of audio, eliminated hallucination entirely. For other low-resource languages written in scripts that are sparsely represented in Whisper's training data, an inspection of the output script therefore constitutes an inexpensive preliminary diagnostic that should precede the reporting of WER.

Sources of improvement. Once the script confusion was resolved, the principal sources of improvement were adaptation capacity and training data. Adapting every linear layer, including the feed-forward layers, substantially outperformed adapting only two attention matrices, which suggests that adaptation to Sindhi requires changes beyond the attention mechanism. Beam search proved beneficial only after fine-tuning; applied to the unadapted model, it reinforced confident but erroneous output, a consideration of relevance to any study reporting zero-shot baselines with beam search.

Data quantity versus model size. At 500 utterances, whisper-large-v3-turbo outperformed whisper-small by 6.1 WER points, yet doubling the training data for whisper-small produced a larger gain. For communities developing speech technology for Sindhi, the collection and validation of additional recordings is therefore likely to yield greater returns than the adoption of larger models. The results further indicate that the unvalidated portion of Common Voice is usable as training material, since all training utterances in this study were drawn from it.

Sensitivity to evaluation choices. The same preliminary recipe yields either 72.97% or 44.0% WER depending on the corpus release, the decoding limit and the normalization procedure. Punctuation alone accounts for 6.3 WER points for a well-adapted model, and generic normalizers may delete genuine words such as ڄ and ڃ. It is accordingly recommended that Sindhi speech recognition studies report CER alongside WER, specify the normalization procedure, and ensure that test sentences are excluded from training data.

Nature of the residual errors. Approximately two-thirds of the residual errors of the best model are misspellings of one or two characters, and many whole-word errors involve proper nouns or loanwords. This distribution explains the disparity between CER (6.4%) and WER (18.7%), since a single incorrect character renders an entire word incorrect under WER. The integration of a Sindhi language model or a lexicon of proper nouns may address a considerable proportion of these errors.

7. Limitations and future work

The study was conducted by a single author using free-tier computational resources, and its limitations reflect these constraints.

- **Single training run per configuration.** Each configuration was trained once without confidence intervals, so differences of one to two points may not persist across repeated runs.
- **Limited evaluation sets.** The test set comprises all 328 validated utterances in the release, and the development set comprises 150 unvalidated utterances.
- **Read speech only.** Conversational speech, background noise and regional accent variation were not evaluated.
- **Configuration selected on one model size.** Whisper-medium and whisper-large-v3-turbo were trained only with the configuration identified on whisper-small, and only with 500 utterances.
- **Label quality.** Training utterances were drawn from the unvalidated pool, and three test references contain Latin characters.
- **Rule-based error analysis.** The error categories derive from deterministic rules rather than human annotation and cannot assess intelligibility.

Future work should train on a substantially larger portion of the 60,000-utterance pool, particularly with whisper-large-v3-turbo; repeat key experiments across multiple seeds with bootstrap confidence intervals; obtain error annotations and intelligibility judgements from literate Sindhi readers; extend evaluation to Devanagari-script Sindhi and spontaneous speech; and assess whether a Sindhi language model or proper-noun lexicon reduces the remaining near-miss errors.

8. Conclusion

Unadapted Whisper does not produce written Sindhi: when forced to decode in the language, it generated the correct script for only 3% of utterances and attained a WER of 142.5%. LoRA fine-tuning on a few hundred utterances resolved the script confusion and eliminated hallucination. Applying LoRA to all linear layers and decoding with beam search reduced the WER of whisper-small to 32.0% with 500 utterances, and training on 2,000 utterances reduced it further to 18.7% WER and 6.4% CER. Larger models offered modest improvements, whereas additional training data offered substantial ones. The majority of residual errors are minor misspellings, and reliable evaluation of Sindhi requires a normalizer that preserves ڄ and ڃ as well as data partitions that exclude test sentences from training.

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Data and code availability

The data are drawn from Mozilla Common Voice Scripted Speech 27.0 (Sindhi), available through the Mozilla Data Collective under its terms of use. The model from the preliminary study is publicly available on Hugging Face as aahanlulla18/whisper-medium-sindhi-lora. Notebooks, per-run predictions and the LoRA adapters produced in this study are available from the author upon request.

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